

Timelike entanglement entropy for the double-scaled SYK model

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ABSTRACT: abstract

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1 Introduction

- Timelike EE as potential holographic probe of dS
- One proposed measure for timelike EE is spacetime density kernel
- Goal is to find holographic interpretation. Therefore, define kernel for DSSYK
- DSSYK in triple scaled regime behaves as AdS JT, while also exhibiting dS features
- It is natural to extend kernel to DSSYK

2 Review

Given a Hilbert space $\mathcal{H}$ with density matrix ρ , time evolution $U_t = e^{-iHt}$, and a bipartition $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ with matrix unit bases $\hat{E}_{ij}^A = |i\rangle\langle j| \otimes \mathbb{1}_B$ and $\hat{E}_{kl}^B = \mathbb{1}_A \otimes |k\rangle\langle l|$,

2.1 Definition and properties of the spacetime density kernel

For two subregions in our Hilbert space we introduce the generalized spacetime density kernel GSDK T_{AB} as the timelike entanglement equivalent to the normal density matrix ρ . The kernel encodes all time dependency of our theory and for the two-time correlators we can write

$$\text{Tr} [T_{AB}(t)(O_A \otimes O_B)] = \text{Tr} \left[\rho O_A U_t^\dagger O_B U_t \right]. \quad (2.1)$$

Furthermore it is convenient to define the components of T_{AB} which we will call the response tensor

$$T_{AB}(t) = \sum_{i,j,k,l} C_{ij;kl}(t) E_{ji}^A \otimes E_{lk}^B. \quad (2.2)$$

In this way the two-time correlator from (2.1) becomes a simple sum over the operator components

$$\mathcal{L}_t(O_A, O_B) = \text{Tr} \left[\rho O_A U_t^\dagger O_B U_t \right] = \sum_{i,j,k,l} C_{ij;kl}(t) O_{ij}^{(A)} O_{kl}^{(B)}, \quad (2.3)$$

where $O_{ij}^{(A)} = \langle i | O_A | j \rangle$ and $O_{kl}^{(B)} = \langle k | O_B | l \rangle$ are the matrix elements in the chosen basis. We calculate the response tensor as the trace

$$C_{ij;kl}(t) = \text{Tr} \left[\rho \hat{E}_{ij}^A U_t^\dagger \hat{E}_{kl}^B U_t \right]. \quad (2.4)$$

2.2 Overview of the double-scaled SYK model

In this section the basic properties of the double scaled SYK model are reviewed and our conventions which mainly follow [1] are stated. The SYK model consists of N Majorana fermions which are coupled by a random parameter J in an p -body interaction.

$$H_{SYK} = i^{p/2} \sum_{i_1 \dots i_p} J_{i_1 \dots i_p} \psi_{i_1} \dots \psi_{i_p}. \quad (2.5)$$

In the double scaling limit the model is exactly solvable. We take

$$N \rightarrow \infty \quad p \rightarrow \infty, \quad (2.6)$$

while keeping

$$\lambda = \frac{2p^2}{N} = \text{const.} \quad (2.7)$$

constant. We also define the parameter $q = e^{-\lambda}$.

By summing over chord diagrams we can simplify the problem to be described by the transfer matrix

$$T = \begin{pmatrix} 0 & \frac{1-q}{1-q} & 0 & 0 & \dots \\ 1 & 0 & \frac{1-q^2}{1-q} & 0 & \\ 0 & 1 & 0 & \frac{1-q^3}{1-q} & \\ 0 & 0 & 1 & 0 & \\ \vdots & & & & \ddots \end{pmatrix}. \quad (2.8)$$

In the double scaling limit one can calculate all the moments of the Hamiltonian $\langle \text{Tr}(H_{\text{SYK}})^k \rangle_J$ by exchanging the ensemble averaged traces in the theory by the vacuum expectation values of powers of the transfer matrix T

$$\langle \text{Tr}(H_{\text{SYK}})^k \rangle_J = \langle 0|T|0 \rangle. \quad (2.9)$$

A closed expression for eigenvectors and eigen-energies of the transfer matrix T have been found. In the case of the infinite chord number hilbert space we can parametrize the eigenvalues of T by a continuous parameter $\theta \in [0, \pi]$ and find

$$E(\theta) = -\frac{2\cos(\theta)}{\sqrt{1-q}} \quad (2.10)$$

with eigenvectors consisting of q -deformed Hermite polynomials. We will use the convention

$$T|\theta\rangle = E(\theta)|\theta\rangle \quad (2.11)$$

with the overlap between chord number and energy Hilbert space

$$\langle n|\theta\rangle = \frac{H_n(\cos(\theta)|q)}{\sqrt{(q;q)_n}}, \quad (2.12)$$

where $H_n(x|q)$ denote the q -deformed Hermite-polynomials. Appendix???? The identity operator in the energy space is taking the form of the integral

$$\mathbb{1} = \int_0^\pi d\theta \mu(\theta) |\theta\rangle\langle\theta| \quad (2.13)$$

The integration measure $\mu(\theta)$ is included to ensure normalization between the eigenstates and takes the form

$$\mu(\theta) = \frac{(q, e^{\pm 2i\theta}; q)_\infty}{2\pi}. \quad (2.14)$$

Here the convention is used that $(a_1, a_2, \dots, a_n; q)_\infty = (a_1; 1)_\infty \cdot (a_2; 1)_\infty \cdot \dots \cdot (a_n; 1)_\infty$ and the $f(\pm\theta) = f(\theta) \cdot f(-\theta)$.

The matter correlators take the form of a double integral:

$$\langle \mathcal{O}(t)\mathcal{O} \rangle = \int_0^\pi d\theta d\phi \mu(\theta)\mu(\phi) e^{it(E(\theta)-E(\phi))} \frac{(q^{2\Delta}; q)_\infty}{(q^\Delta e^{i(\pm\theta \pm \phi)}; q)_\infty}. \quad (2.15)$$

3 A spacetime density kernel for the DSSYK model

With the above we can find a response tensor $C_{ij;kl}$ for the double scaled SYK. Here we use the special case that for the derivation of the kernel components we use the full system as system A and system B . So for the basis elements in $C_{ij;kl}$ we use a trivial 1 in the complement tensor space of our system.

This gives us the full chord number basis $\hat{E}_{ij}^A = |i\rangle\langle j|$ and $\hat{E}_{kl}^B = |k\rangle\langle l|$ for the basis states used in eq. (2.4). We will afterwards sum over a subset of components to define the spacetime density kernel T_{AB} .

3.1 Derivation of the DSSYK spacetime density kernel

For the derivation of the response tensor components we can use the well known solution of the SYK model in the double scaling limit. With the solutions from section 2.2, it is straightforward to calculate eq. (2.4) for $\{i, j, k, l\}$ being states in the chord number space. We can also exchange the trace and averaging by a simple vacuum expectation value in the auxiliary chord Hilbert space. After this simplifications the response tensor takes the form

$$C_{ij;kl} = \langle 0 | \rho | i \rangle \langle j | e^{iTt} | k \rangle \langle l | e^{-iTt} | 0 \rangle \quad (3.1)$$

By inserting the identity operator in the energy space representation twice, we reach a closed integral form in terms of

$$C_{ij;kl}(t) = \langle 0 | \rho | i \rangle \int \frac{d\theta d\phi}{(2\pi)^2} \mu(\theta) \mu(\phi) e^{-it(E(\theta) - E(\phi))} \cdot \frac{H_j(\cos \theta | q) H_k(\cos \phi | q)}{\sqrt{(q; q)_j (q; q)_k}} \frac{H_l(\cos \theta | q)}{\sqrt{(q; q)_l}}. \quad (3.2)$$

With this we have defined an explicit form of the components of the generalized spacetime density kernel, which we can use to calculate timelike entanglement measures for the double scaling SYK model. In the following we will show that the kernel indeed behaves as we expect.

3.2 Relation to the two-point function

From the definition of the kernel [ref?](#) we expect that a simple sum of the kernel components with the components of the operators in the appropriate basis gives us back the two time correlation function of this operator. In DSSYK we can write the components of the operators as a diagonal matrix in the chord space. The components of the matrix $S = \text{diag}(1, q^\Delta, q^{2\Delta}, \dots)$ go into eq. (2.3), and we choose the chord vacuum state $\rho_0 = |0\rangle\langle 0|$ as reference state. From this assumptions we get

$$\mathcal{L}_t(\mathcal{O}, \mathcal{O}) = \sum_{ijkl=0}^{\infty} \langle 0 | i \rangle \int \frac{d\theta d\phi}{(2\pi)^2} \mu(\theta) \mu(\phi) e^{-it(E(\theta) - E(\phi))} \cdot \frac{H_j(\cos \theta | q) H_k(\cos \phi | q)}{\sqrt{(q; q)_j (q; q)_k}} \frac{H_l(\cos \theta | q)}{\sqrt{(q; q)_l}} \delta_{ij} \delta_{kl} q^{i\Delta} q^{k\Delta}. \quad (3.3)$$

With the properties of the q-Hermite polynomials [Appendix ref!!!](#), we have $H_0(x|q) = 1$, $(q; q)_0 = 1$ and are left with a sum over two q-Hermite polynomials.

$$\mathcal{L}_t(\mathcal{O}, \mathcal{O}) = \int \frac{d\theta d\phi}{(2\pi)^2} \mu(\theta) \mu(\phi) e^{-it(E(\theta) - E(\phi))} \cdot \sum_{k=0}^{\infty} \frac{H_k(\cos \phi | q) H_k(\cos \theta | q)}{(q; q)_k} q^{k\Delta}. \quad (3.4)$$

Here we can use the orthogonality relations 5.1 and find the known form of the two point function eq. (2.15) and can confirm

$$\sum_{i,j,k,l} C_{ij;kl}(t) \mathcal{O}_{ij} \mathcal{O}_{kl} = \int_0^\pi d\theta d\phi \mu(\theta) \mu(\phi) e^{it(E(\theta) - E(\phi))} \frac{(q^{2\Delta}; q)_\infty}{(q^\Delta e^{i(\pm\theta \pm \phi)}; q)_\infty} \quad (3.5)$$

for our DSSYK version of the kernel.

3.3 Spectral form factor from kernel

4 Numerical results

4.1 Timelike entanglement measures

The spacetime density kernel allows us to study correlations in time, since it encodes the dynamics of different time-slices in one operator. Multiple measures, which quantify these correlations, can be extracted from the spacetime density kernel $T_{AB}(t)$. The goal of this section is to numerically calculate these quantities, which were introduced in [2], for the double-scaled SYK model. It is important to note that $T_{AB}(t)$ is not necessarily Hermitian, but $\text{Tr } T_{AB}(t) = 1$ always holds. Firstly, we aim to calculate the Tsallis entropy

$$Z_n(t) = \text{Tr}[T_{AB}(t)^n], \quad (4.1)$$

while we mostly focus on the $n = 2$ case. We also investigate the Schatten p -norms of T_{AB} , which are defined by

$$\|M\|_p = \text{Tr} \left((M^\dagger M)^{p/2} \right)^{1/p}. \quad (4.2)$$

In particular, we are interested in the Frobenius norm

$$\|T_{AB}(t)\|_2^2 = \text{Tr} \left[T_{AB}(t) T_{AB}^\dagger(t) \right]. \quad (4.3)$$

In order to quantify the non-Hermiticity of the spacetime density kernel, we also study the entanglement p -imagitivity

$$\mathcal{I}_p(t) = \|T_{AB}(t) - T_{AB}^\dagger(t)\|_p. \quad (4.4)$$

Here, we focus again on the case $p = 2$, since $\mathcal{I}_2(t)$ controls causal influence: it is nonzero iff signaling between A and B is possible.

Another important quantity which we will consider is the kernel negativity $\mathcal{N}_H(t)$, which quantifies the spectral weight of the Hermitian part $H_{AB}(t) \equiv (T_{AB} + T_{AB}^\dagger)/2$ of the spacetime density kernel. It is defined by

$$\mathcal{N}_H(t) = \sum_{\mu_\alpha(t) < 0} |\mu_\alpha(t)|, \quad (4.5)$$

where $\{\mu_\alpha(t)\}$ are the eigenvalues of $H_{AB}(t)$.

4.2 Spectral form factor

5 Discussion and outlook

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Appendix

$$\begin{aligned} \sum_{n=0}^{\infty} H_n(x|q) H_n(y|q) \frac{t^n}{(q; q)_n} &= \frac{(t^2; q)_{\infty}}{(te^{i(\theta+\phi)}, te^{i(\theta-\phi)}, te^{-i(\theta+\phi)}, te^{-i(\theta-\phi)}; q)_{\infty}} \\ &= \frac{(t^2; q)_{\infty}}{(te^{i(\pm\theta\pm\phi)}; q)_{\infty}} \end{aligned} \tag{5.1}$$

References

- [1] M. Berkooz, M. Isachenkov, V. Narovlansky and G. Torrents, *Towards a full solution of the large N double-scaled SYK model*, *JHEP* **03** (2019) 079 [[1811.02584](#)].
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